MIL-C-17317A(SHIPS)
24 March 1967
SUPERSEDING
MIL-G-17317(SHIPS)
30 August 1952
(See 6.3)

MILITARY SPECIFICATION

COMPUTER, SHIP'S STABILITY

1. SCOPE

1.1 This specification covers a stability computer, a calculating device which automatically predetermines the mean draft, displacement, deadweight and stability which a ship will have under any condition of loading or at any time of a voyage.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-983 - Interior Communication Equipment; Naval Shipboard, Basic Design Requirements for.
MIL-M-15071 - Manuals, Equipment and Systems.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 10016.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. - Prior to beginning production, a sample shall be tested as specified in 4.2 (see 6.2).

3.2 General features. - The equipment shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein. (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern).

General requirements
Definitions
Materials, general
Substitution of (equal or superior) materials or parts
Springs (material)
Other metals

FSC 6605

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Plastics
Ceramics
Glass
Lubricants and lubrication
Painting (see 3.5)
Protection against corrosion
Bolts, machine screws, studs and nuts
Parts - mechanical
Locking devices
Washers
Bearings, ball
Enclosures - general
Enclosure - accessibility
Enclosure - degree of
Enclosure - mounting
Stiffening grooves
Threaded devices
Rounded corners and edges
Drilled and tapped holes
Structural welding
Temperature and humidity
Inclination
Drawings - general
Drawings - preliminary
Drawings - working
Bill of materials
Designation and marking (Serial numbers are required)
Item names and nomenclature
Workmanship

3.3 Design features.-

3.3.1 The computer shall be as compact and lightweight as practicable and shall be suitable for desk or portable use aboard ship or ashore. A handle shall be provided on one of the sides or top of the case for the purpose of carrying the instrument.

3.3.2 Reliability.- The reliability of the computer must meet the requirements of Naval service. It is imperative that reliability of operation be considered of prime importance in the design and manufacture of the computer. The contractor shall employ all methods possible in the process of manufacture which will assure quality and maximum reliability consistent with the state of the art. The equipment shall represent the simplest design consistent with the functional and environmental requirements specified herein.

3.3.3 Electrical power.- No electrical power shall be required to operate the computer. The computer shall be completely independent of internal or external electrical power and the displayed information shall be clearly visible and readily interpreted from a distance of 18 inches.

3.3.4 Operation of the computer shall be made as simple as practicable and shall be suited for use by nontechnical personnel.

3.3.5 Inputs to the computer shall be the total loadings in tons at various deck levels or heights, and corrections for free surfaces of liquids or grain and nonhomogeneous loads. The input actuators shall be set and displayed so that they jointly form a status board of the vertical load distribution.

3.3.6 Inputs shall be manually set into the computer by means of micrometer thimbles. Separate thimbles shall be provided for each input. The load inputs shall be located on the right side of the computer.

3.3.7 The scale setting for the load input thimbles shall correspond to the total load or deadweight stowed on the respective deck level.

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3.3.8 The computer shall compute and indicate automatically the following outputs:

(a) Total load or deadweight in tons.
(b) Total displacement in tons.
(c) Mean draft in feet.
(d) "GM" in feet (distance between ship's-metacenter and the loaded ship's center of gravity).
(e) Limiting stability.

3.3.9 The limiting stability condition shall be indicated by a red warning flag appearing in the window of the "GM" scale. This warning flag shall appear whenever a load is contemplated which does not provide the minimum stability required.

3.3.10 Calibration.- The output scales shall be determined from information to contractor such as Capacity Plan, Curves of Form and Stability Curves, as applicable, to the specific type of vessel and other data required by the contractor to design the computer.

3.3.10.1 Scale markings and numerals.- The markings and numerals shall be black on a white background.

3.3.11 Accuracy.- When used according to instructions, the computer shall indicate:

(a) "GM" to within $\pm$ 0.2 feet.
(b) The weight of cargo within $\pm$ 100 tons for ships up to 9,000 tons deadweight, from 9,000 to 12,000 tons $\pm$ 150 tons and from 12,000 to 16,000 tons $\pm$ 200 tons.

3.4 Operating.- An operating test shall be conducted for each computer as specified in 4.4.

3.5 Painting.- The front panel shall be of white core bakelite with lines and letters engraved and filled with suitable paint. The case shall be painted with a grey finish. The name or type of vessel shall appear on the front panel of each instrument.

3.6 Manuals.- Manuals shall be type I of MIL-M-15071.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection.- Preproduction inspection shall consist of the examination and tests specified in table I which shall be performed, in general, in the order listed.

Table I - Preproduction inspection.

Inspection	Requirement paragraph	Test reference
General examination	---	MIL-I-983
Operating	3.4	4.4
Temperature and humidity	3.2	MIL-I-983
Inclination	3.2	MIL-I-983
Enclosure	3.2	MIL-I-983

4.3 Quality conformance inspection.- All computers offered for delivery on the contract or order shall be subjected to the examination and test specified in table II. Any equipment failing to meet the requirements of this specification shall not be offered for delivery.

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Table II - Quality conformance inspection.

Inspection	Requirement paragraph	Test reference
General examination	---	MIL-I-983
Operating	3.4	4.4

4.4 Operating test. - The operating test shall consist of five bench trials each simulating a hypothetical loading operation. The computer shall indicate automatically the outputs specified in 3.3.8 with accuracy as specified in 3.3.11. Tolerance deviation in excess of tolerances specified in 3.3.11 must be recorded and subsequently resolved to the satisfaction of the cognizant Government inspector.

4.5 Inspection of preparation for delivery. - The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation. -

5.1.1 Computers. -

5.1.1.1 Preservation and packaging. - Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment. - The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation, packaging, packing and marking. - The equipment and accessories, and technical publications shall be preserved and packaged levels A or C; packed level A or B, and marked in accordance with MIL-E-17555.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of items required.
(c) Quantity of manuals required.
(d) Preservation and packaging, packing, and marking if other than as specified in 5.1 (see 5.2).

6.2 Preproduction. - Invitations for bid should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

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6.3 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 6605-N109Sh)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

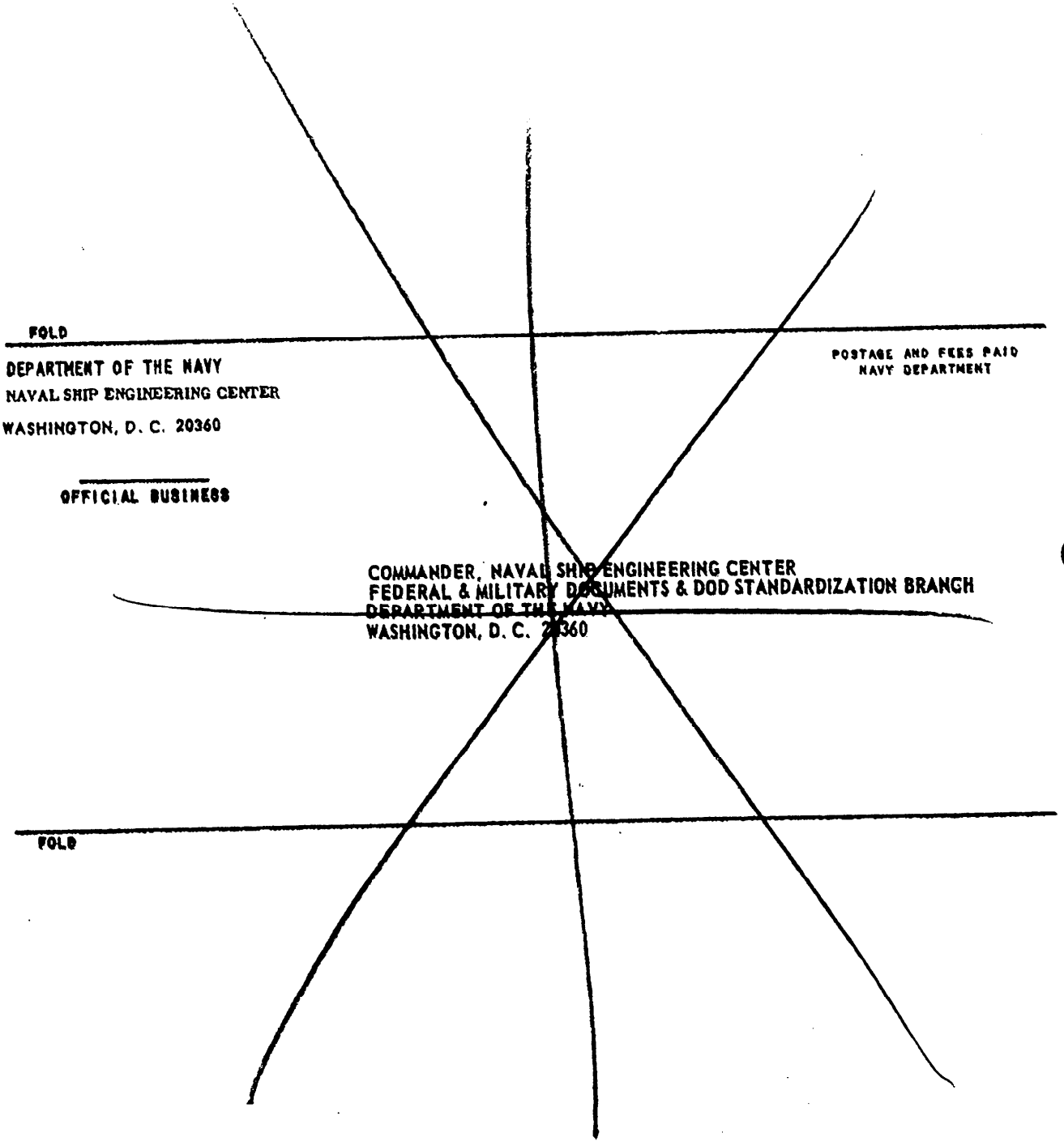
4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

(NAVSHIPS OVPRT 13-66)

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